

Validity Analysis: DRBENCHMARK

Target context: French Hospital Clinical NLP — Metropolitan France (DrBenchmark)

Overall risk: **HIGH**

Deployment Context

Use case: In a hospital setting, a clinician uses software to manage unstructured French clinical cases by applying a model evaluated on part-of-speech tagging, named-entity recognition, and multi-label classification. The system identifies specific clinical and temporal entities while mapping pathologies to standardized medical classification chapters and axes to automatically generate a patient's clinical profile. These outputs directly help the clinician summarize complex medical histories and prioritize care based on the severity of identified symptoms, ensuring that life-critical decisions are supported by accurately synthesized data.

Target population: French-speaking medical professionals, including clinicians, hospital physicians, and nursing staff in French-speaking health systems.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ✓	3	Moderate gaps	high
Input Content △	2	Significant gaps	high
Input Form ✓	4	Minor gaps	high
Output Ontology △	2	Significant gaps	high
Output Content △	2	Significant gaps	high
Output Form △	2	Significant gaps	high
Average	2.5		

△ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

DrBenchmark is the most comprehensive French biomedical NLP benchmark available and provides genuine partial coverage for the Metropolitan French hospital clinical NLP deployment, particularly through DEFT-2021 (multi-label MeSH classification + rich NER with temporal entities), CAS and CLISTER (clinical-case French), and DiaMed (only ICD-10 chapter-level labels

with documented IAA and medical-expert involvement). Input Form is well aligned (text-only standard French, Latin diacritics, HuggingFace toolkit). However, four of the six dimensions show significant validity concerns for this deployment: Input Content (score 2) — no raw-EHR register, substantial non-metropolitan content in DiaMed and MorFITT, register systematically cleaner than active hospital notes; Output Ontology (score 2) — no confidence-scored / ranked-candidate / uncertainty-flagging output structure, ICD-10 chapter granularity coarser than operational PMSI/GHM codes, several datasets with orthogonal labels (negation/speculation, pharmacy MCQA); Output Content (score 2) — no documented PMSI coder or hospital coding specialist involvement in any constituent dataset, automatic silver-standard POS annotations in CAS/ESSAI, annotation completeness concerns in E3C and ESSAI; Output Form (score 2) — standard F1/Spearman/SeqEval cannot measure the deployment's required confidence calibration and uncertainty flagging behavior, severe class imbalance in CAS/ESSAI/PxCorpus masks minority-class performance. Input Ontology scores 3 due to broad task coverage but with several irrelevant categories (pharmacy exam, veterinary, encyclopedic) consuming evaluation weight. The deployment's elicitation-flagged HIGH-priority concerns (IC, OO, OC) all map to validity scores of 2.

Practical Guidance

What This Benchmark Measures

DrBenchmark measures core French biomedical language understanding capabilities — POS tagging, clinical and temporal NER, multi-label clinical-profile classification, semantic textual similarity, and intent classification — across 8 state-of-the-art masked language models with reproducible 4-run averaged metrics. For the deployment, the strongest constructs measured are: (1) French clinical NER for pathology, anatomy, treatment, dosage, and temporal entities (via DEFT-2021's 14-type scheme and E3C temporal annotations); (2) multi-label classification capacity across MeSH and ICD-10 chapter axes (via DEFT-2021 and DiaMed); (3) POS tagging on French clinical case text (via CAS).

Construct Depth

Construct depth is shallow-to-moderate for the deployment's core functions. Input Form (4) and Input Ontology (3) provide reasonable construct coverage, but the four dimensions most consequential to the deployment — Input Content, Output Ontology, Output Content, Output Form — all score 2. The benchmark provides binary or single-label evaluation only; the deployment's required confidence-scored multi-label outputs with uncertainty flagging is structurally unmeasurable. Annotation authority for the deployment-critical ICD-10/PMSI mapping function rests on a single medical expert in one non-metropolitan dataset (DiaMed). Aggregate scores conflate clinically relevant tasks with off-domain content (pharmacy exam, veterinary, encyclopedic STS pairs).

What Else You Need

Three concrete supplementation streams are needed before deployment-relevant evaluation conclusions can be drawn: (1) Custom calibration evaluation — calibration curves, ECE, Brier scores, per-label AUROC, and ranked-candidate quality metrics on DEFT-2021 and DiaMed multi-label outputs (addresses OO/OF gaps); (2) Re-annotation by certified PMSI coders on a held-out metropolitan French hospital subset to validate label authority for ICD-10/PMSI mapping (addresses OC gap and the elicitation-flagged annotator-vs-user gap); (3) Construction or curation of a moderate-noise active-hospital-note evaluation set with French clinical abbreviations (ATCD, TTT, HDJ) and dictation artifacts to assess register-transfer performance (addresses IC gap, since web search confirms no French raw-EHR benchmark exists [WEB-1, WEB-18]). Aggregate DrBenchmark scores should be decomposed task-by-task and filtered to exclude FrenchMedMCQA, MorFITT veterinary content, and DEFT-2020 off-domain encyclopedic pairs when interpreting deployment-relevant signal.

ID	Evidence
WEB-1	EHRNoteQA (NeurIPS 2024) demonstrates that existing clinical NLP benchmarks typically fail to represent raw multi-note EHR documentation; no equivalent French benchmark identified (https://arxiv.org/abs/2402.16040)
WEB-18	Recent 2025 meta-analysis confirms a significant proportion of medical LLM benchmarks are constructed from academic materials rather than authentic clinical data — applicable to DrBenchmark's source composition (https://arxiv.org/html/2508.04325)